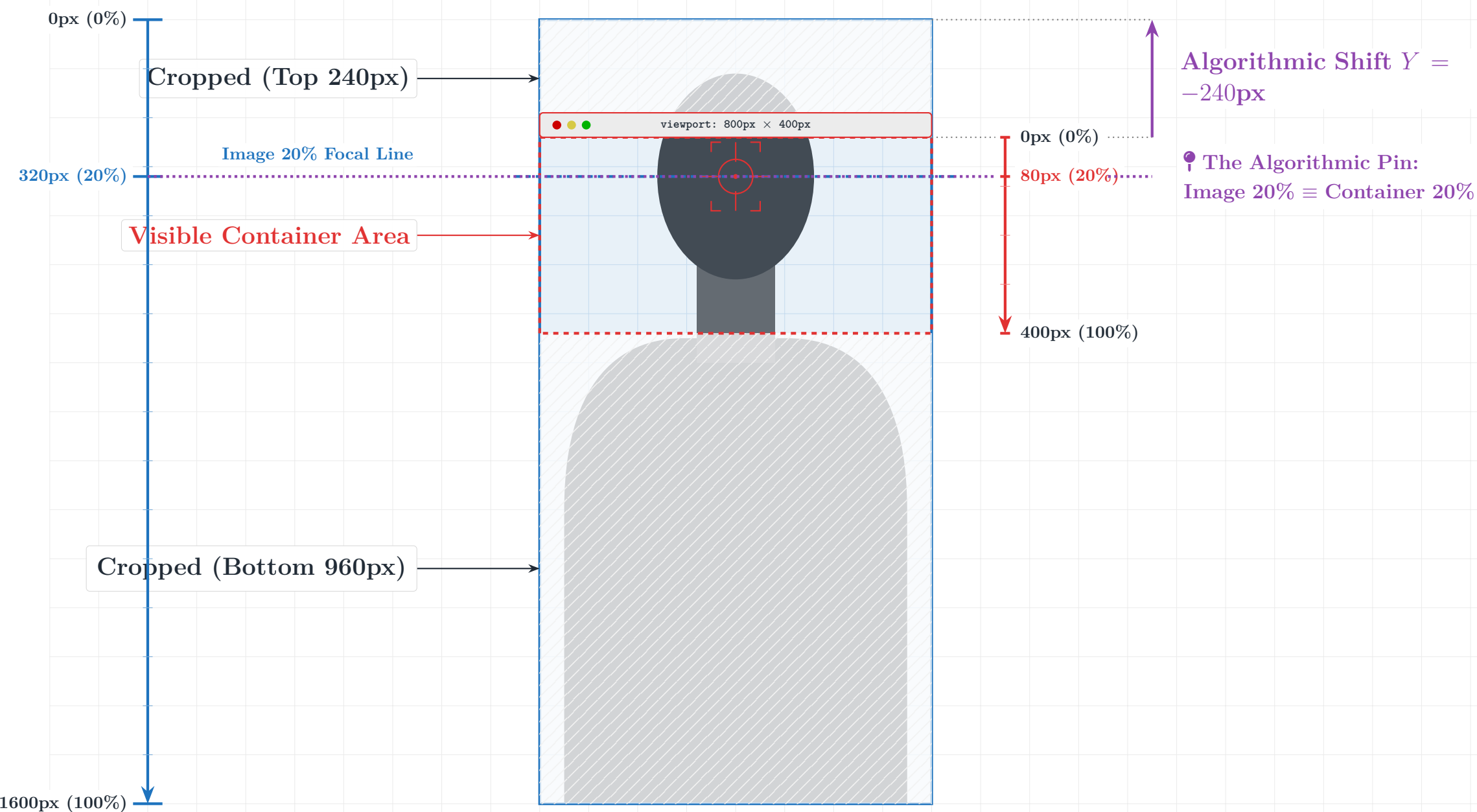


Algorithmic Pinning via object-position

Mapping internal coordinate offsets to lock a focus point (object-position: 50% 20%)



1. Dimensions After cover

The browser scales a portrait image to cover a landscape container, preserving aspect ratio:

- Container Height (H_c): **400px**
- Scaled Image Height (H_i): **1600px**

*This size disparity creates a massive **1200px** structural overflow on the Y-axis.*

2. Identifying 20% Marks

The 20% value in CSS commands the engine to find the 20% Y-coordinate in both local spaces:

- **Image:** $1600\text{px} \times 0.20 = 320\text{px}$
- **Container:** $400\text{px} \times 0.20 = 80\text{px}$

Goal: Translate the image so its internal 320px focal mark physically aligns with the container's 80px mark.

3. The Shift Formula

The absolute physical translation is algebraically calculated by subtracting the focal differences:

$$\begin{aligned}\text{Offset}_Y &= (H_c - H_i) \times \text{Position}_Y\% \\ \text{Offset}_Y &= (400 - 1600) \times 0.20 \\ \text{Offset}_Y &= -1200 \times 0.20 = \mathbf{-240\text{px}}\end{aligned}$$

The CSS rendering engine shifts the image origin **240px UPWARDS**.

4. What about the 50% X-Axis?

Since the image width perfectly matches the container width after scaling ($W_c = W_i = 800\text{px}$):

$$\begin{aligned}\text{Offset}_X &= (800 - 800) \times 0.50 \\ \text{Offset}_X &= \mathbf{0\text{px}}\end{aligned}$$

No horizontal shift occurs; the face remains mathematically locked in the absolute center horizontally.